Objective: Design a 555-based adjustable clock-pulse generator.

C**ircuit operation:** This circuit works as an pulse generator with a controllable frequency of both high & low signal. By using 2 potentiometer, it is possible to control PWM. With a simple diode parallel to R_2 (connected between discharge & trigger), we can adjust the time of either low or high signal without effecting each other. Turning the potentiometer to minimum to maximum resistance can give user the freedom according to their needs.

Theoretical Calculation: The formula to calculate times,

$$T_{max} = 0.693 * R_1 * C_1$$
$$T_{min} = 0.693 * R_2 * C_1$$
$$T_{total} = T_{high} + T_{low}$$
$$F = 1/T_{total}$$

With possible maximum resistance:

$$T_{max/min} = 0.693 * (501*10^3) * (10*10^{-6}) \text{ s}$$
$$= 3.472 \text{ s}$$

So, the frequency is at minimum. Frequency = 0.144 Hz

With possible minimum resistance:

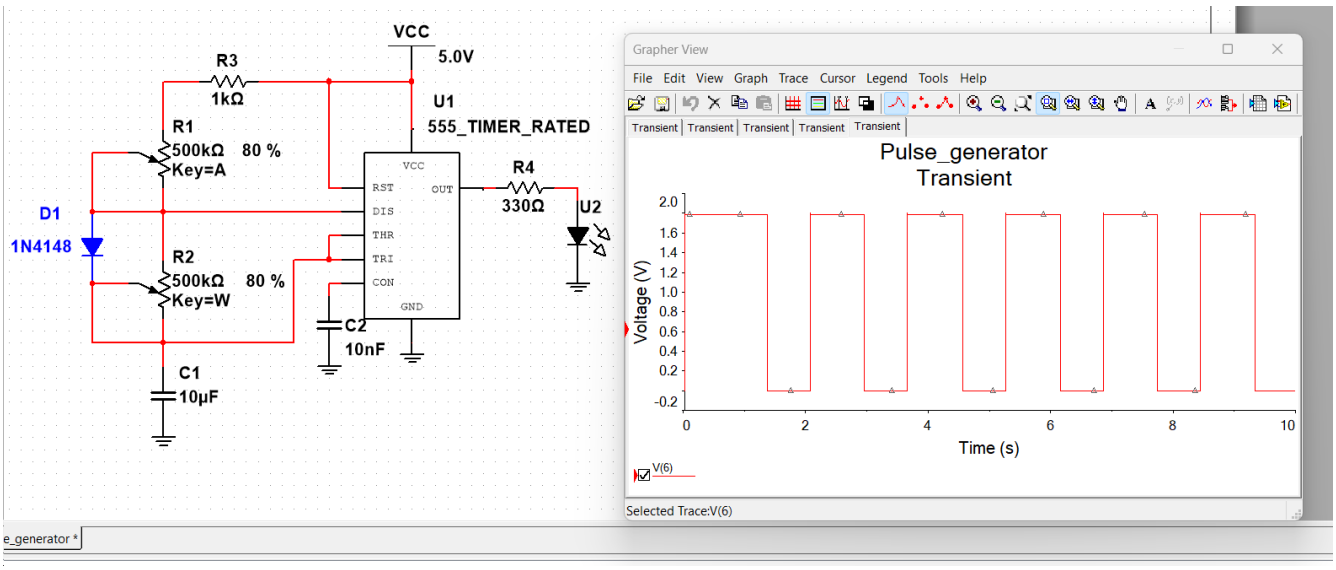
$$T_{max/min} = 0.693 * (1*10^3) * (10*10^{-6}) \text{ s}$$
$$= 0.693 \text{ s}$$

So, the frequency is at maximum. Frequency = 0.722 Hz

Note that all of the calculations were done based on that duty cycle are 50%. Where-

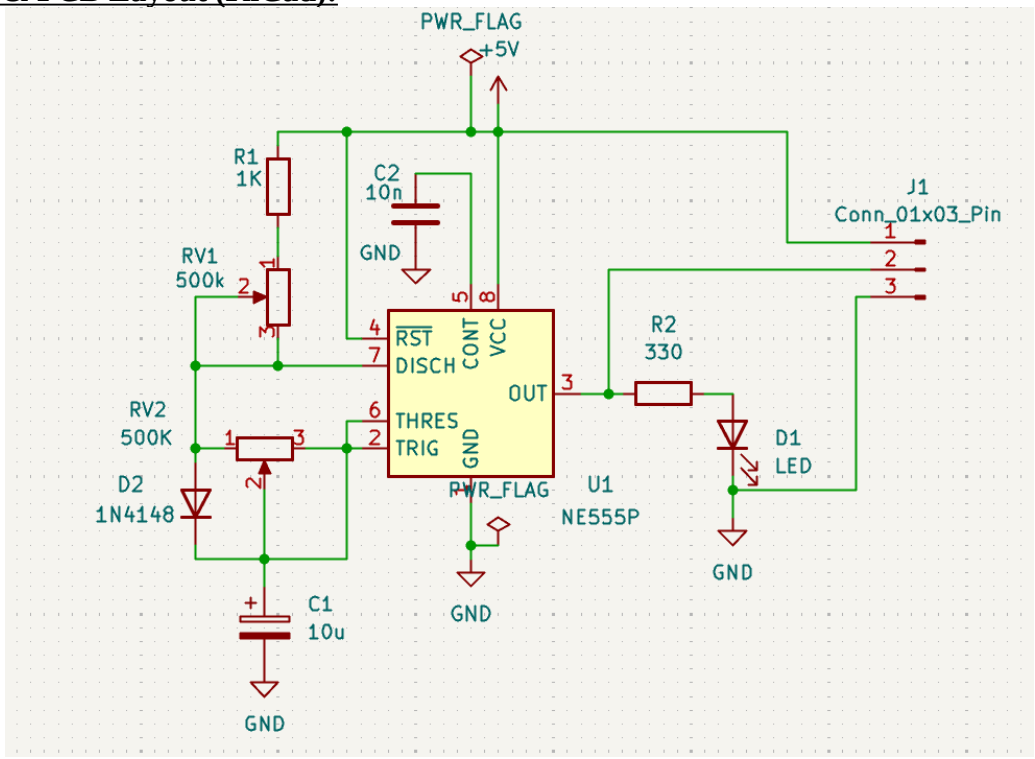
duty cycle% = (Total cycle – High cycle)/Total cycle *100%

Multisim simulation:

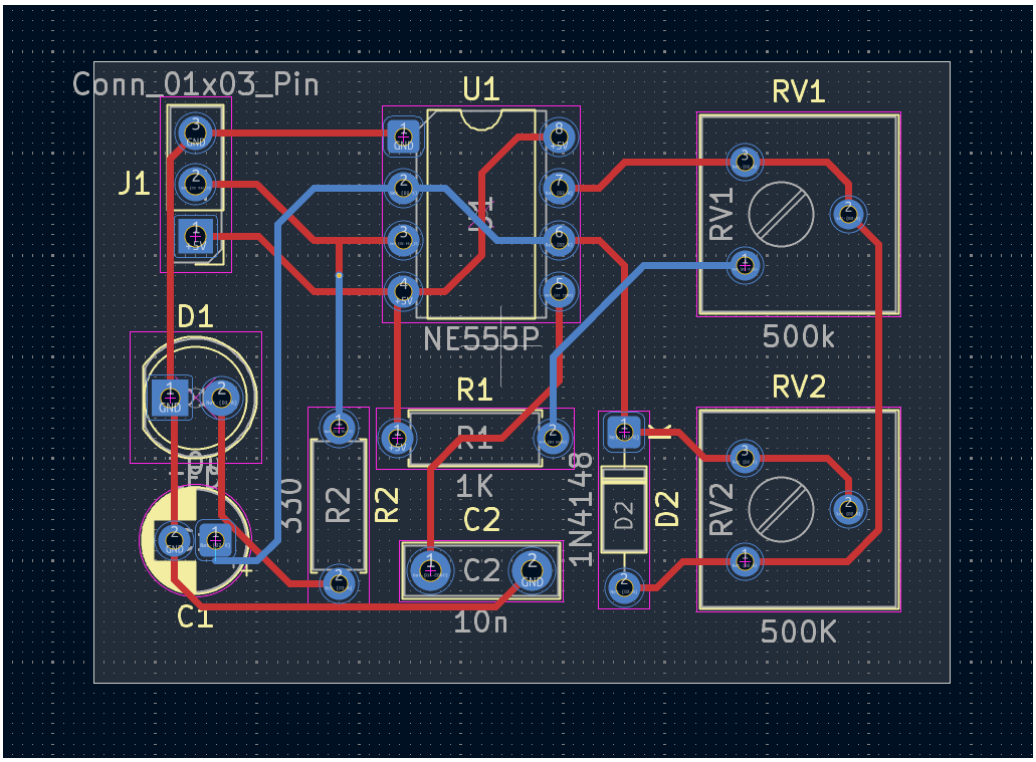


Here R1 & R2 values are 400 ohms & duty cycle is at 50%

Schematic & PCB Layout (KiCad):



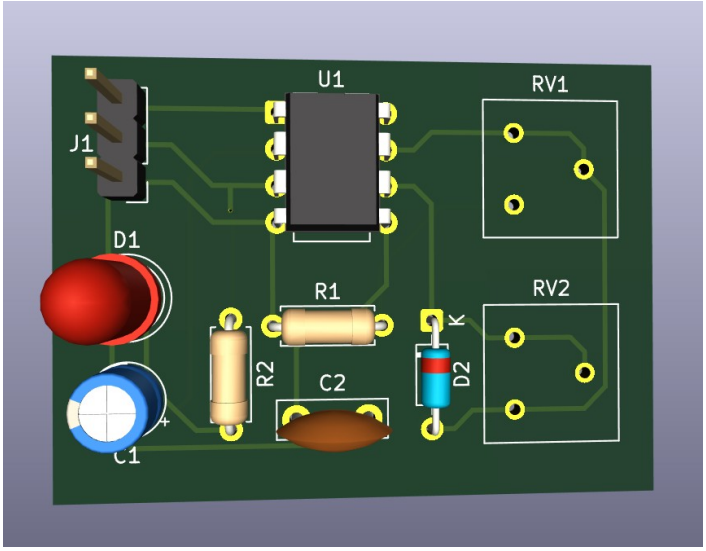
An LED at the end which acts as an indicator for the user. Also added connecting pins for the power supply, output & ground.

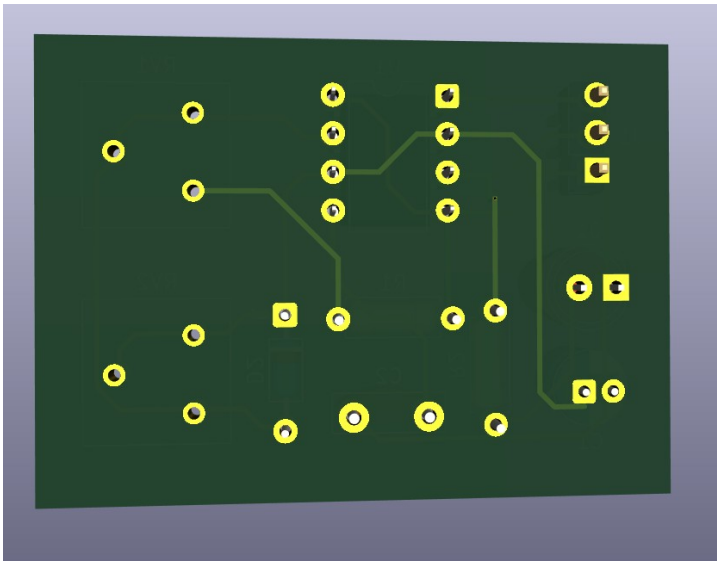


All the grounds are common at the 3rd connector pin. Red lines are front copper traces & Blue lines are

back traces.

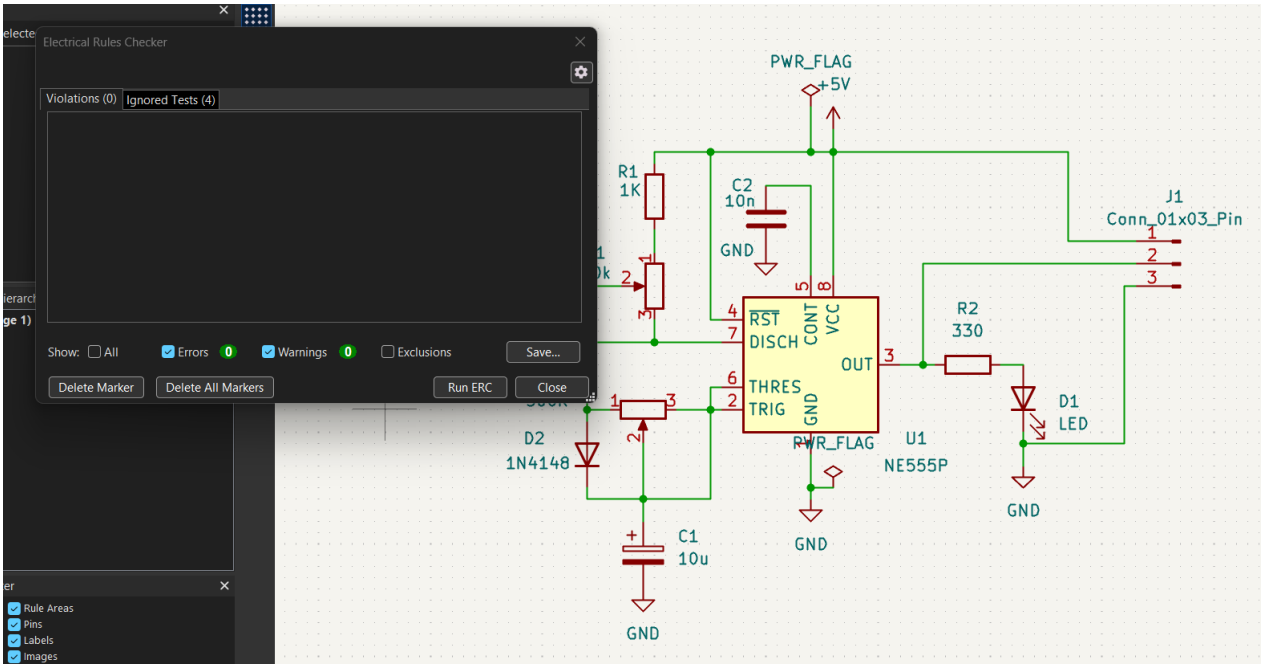
3D Render:





All components & traces are showed in the 3d View. The model for potentiometer were not available.

Schematic Electrical Rules Check :



0 errors & warnings are found from the checkup. A 3D render does not prove that the board is manufacturable or that the circuit works.